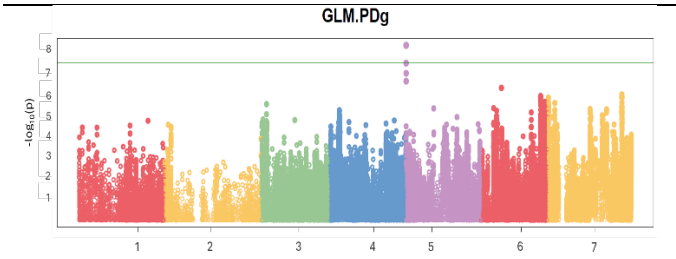
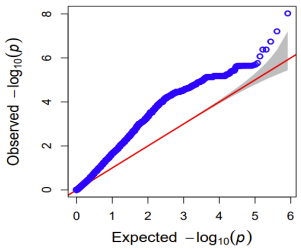
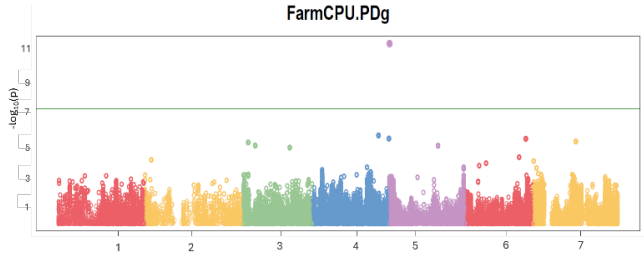
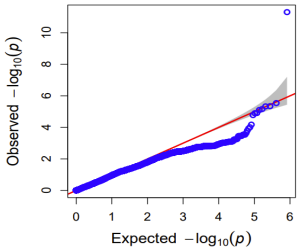
Greenhouse experiment

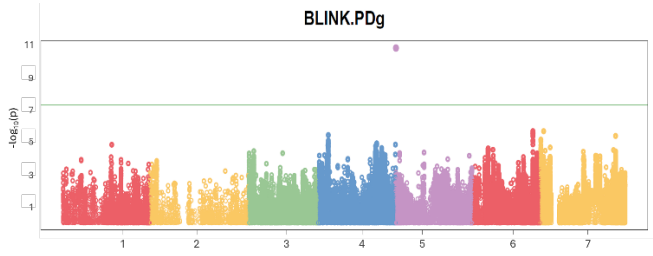
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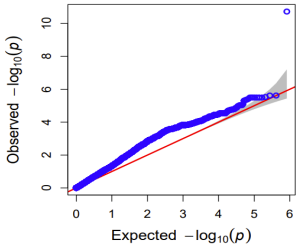








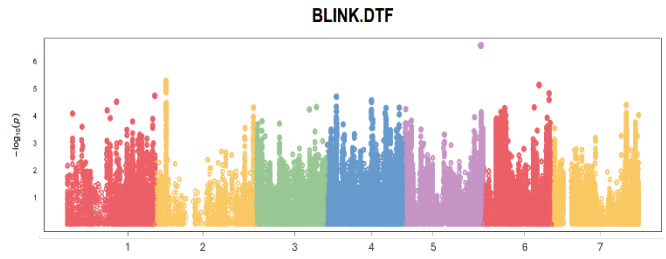


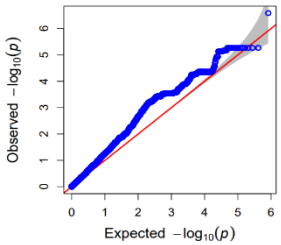


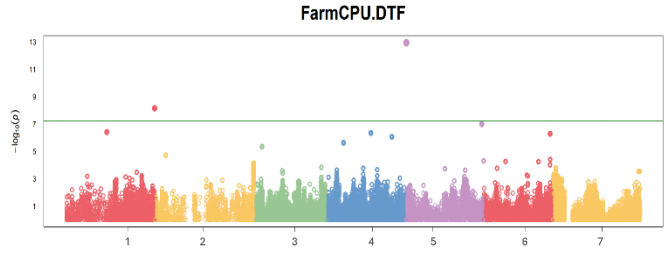
Multi-environment experiment

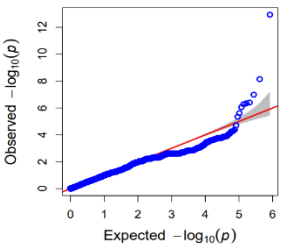
DTF

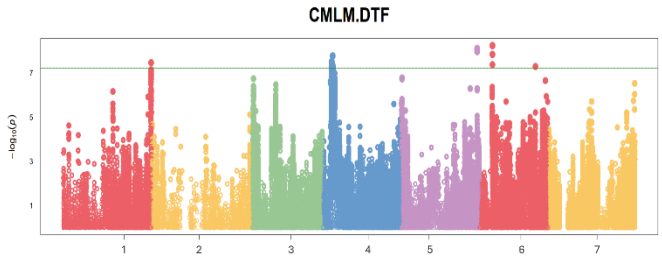
WPR2023

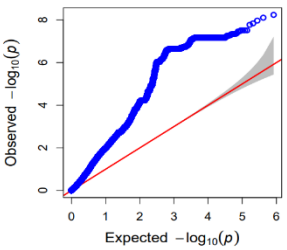


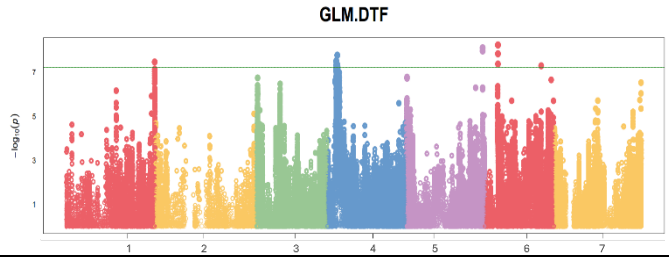


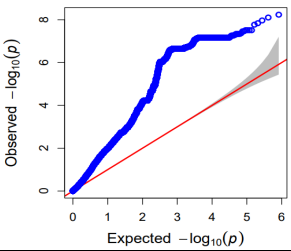




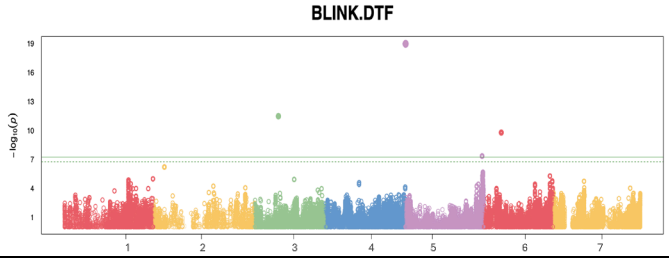


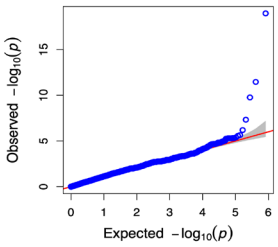


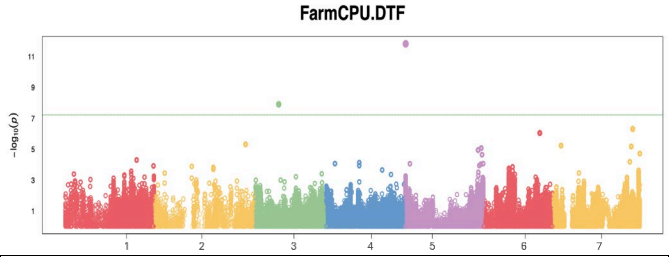


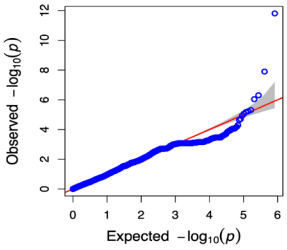


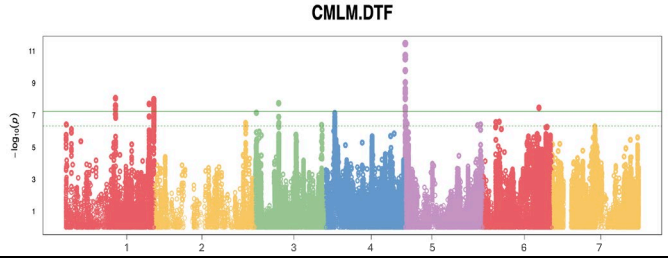
CHE2023

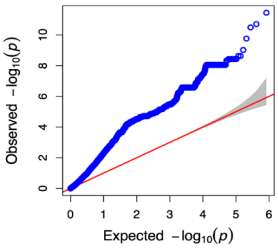


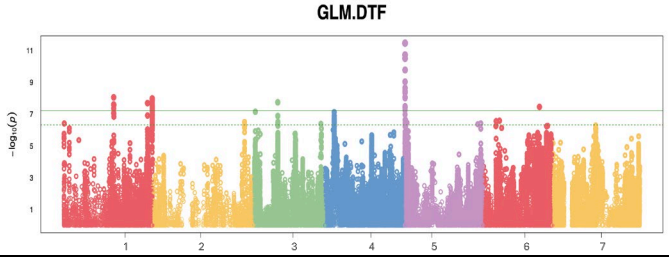


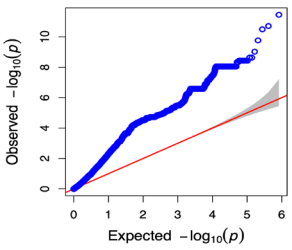




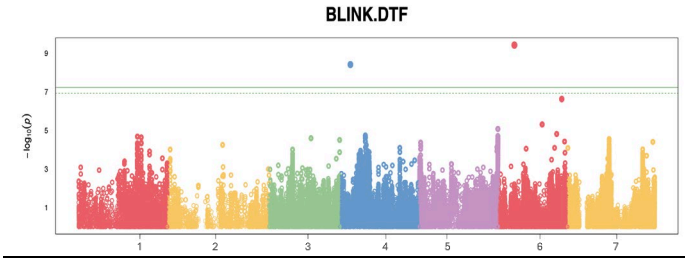


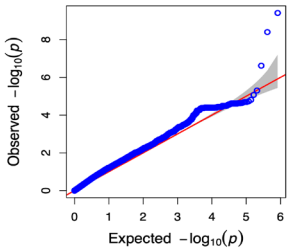


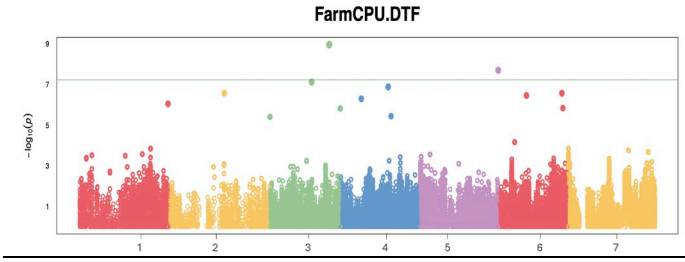


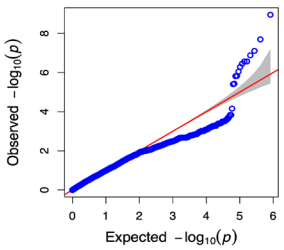


WPR2024

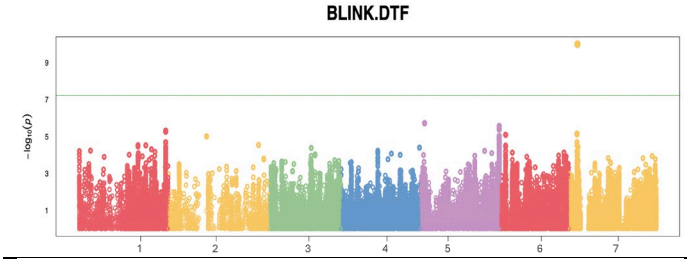


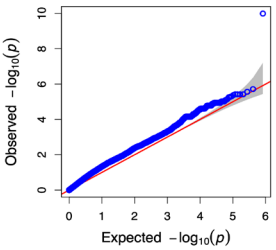


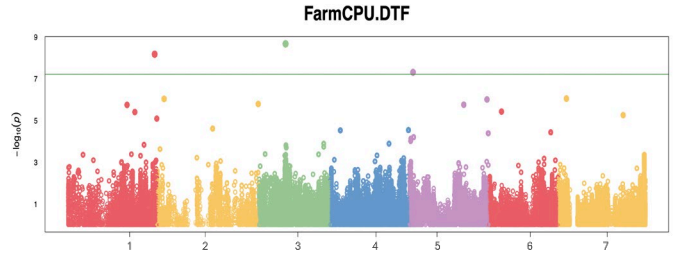


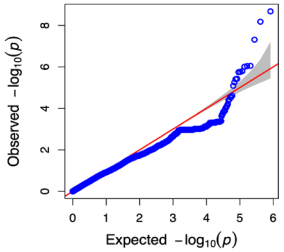


SIM2024



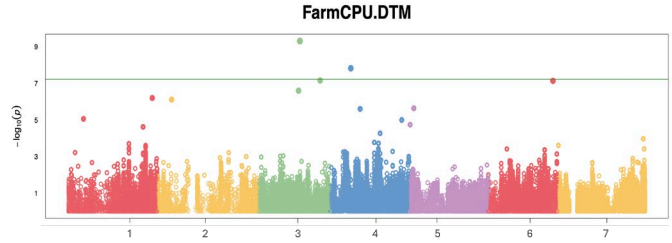


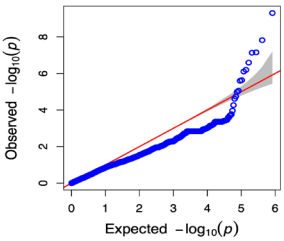


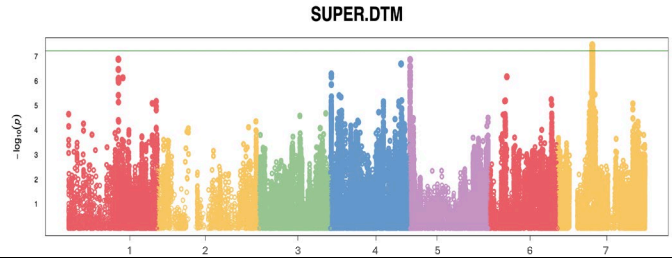


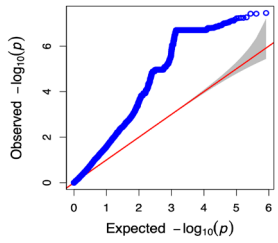
DTM

WPR2022

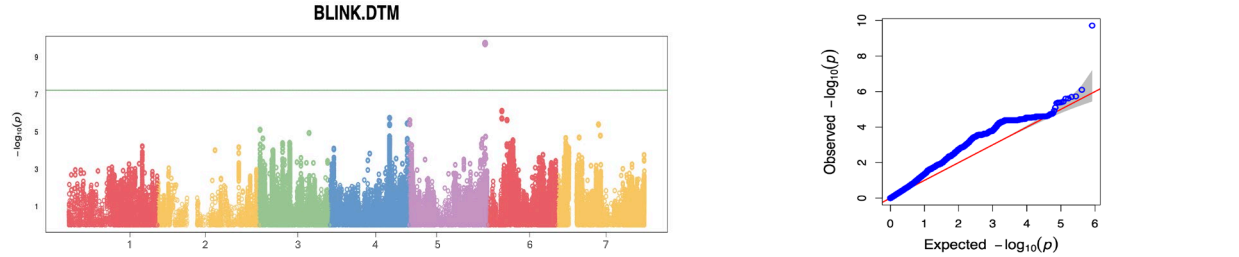


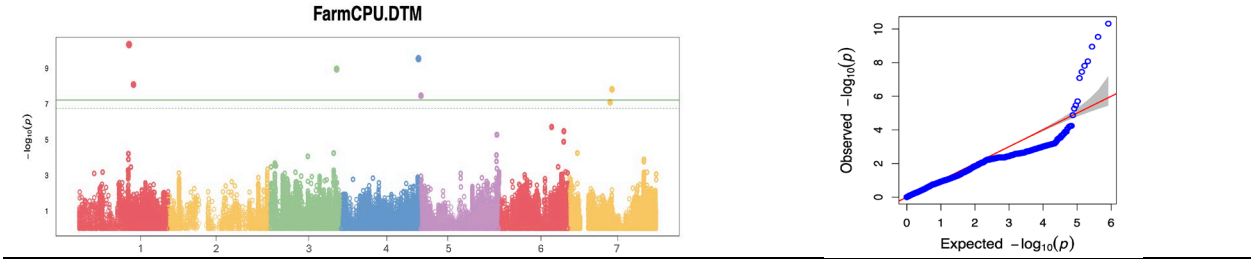


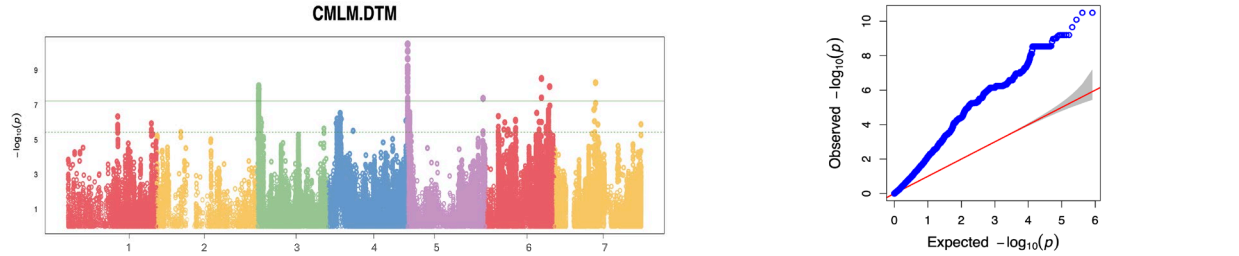


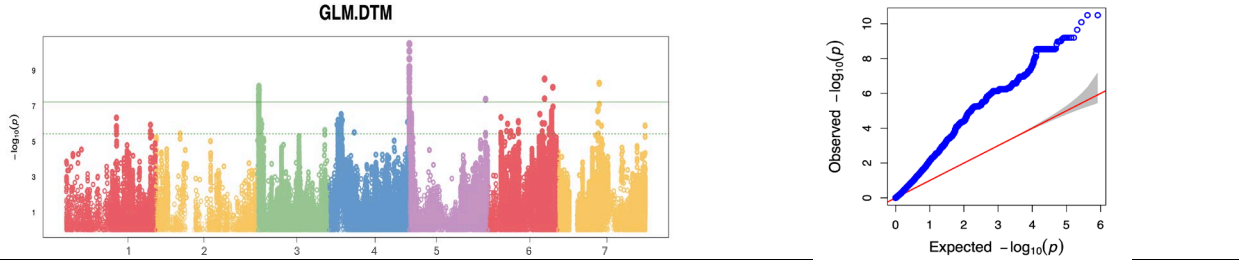


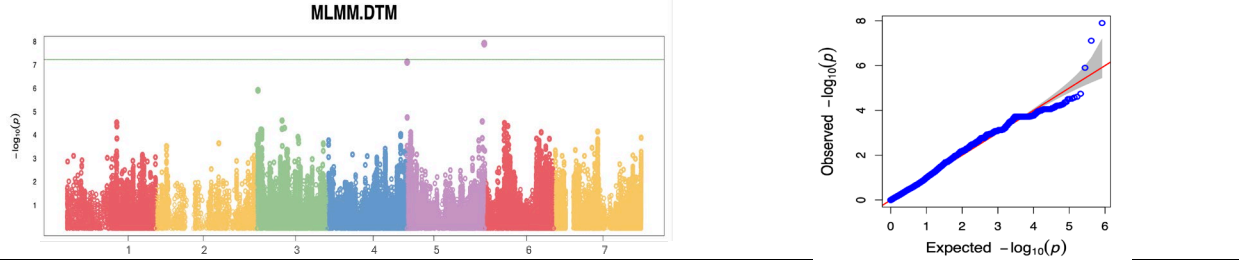
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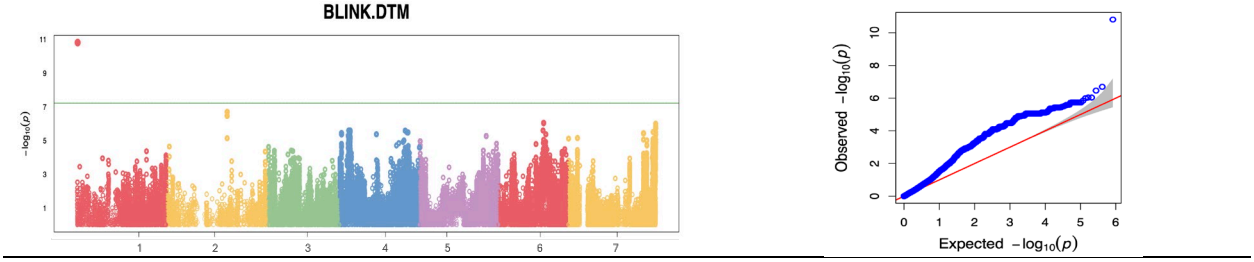




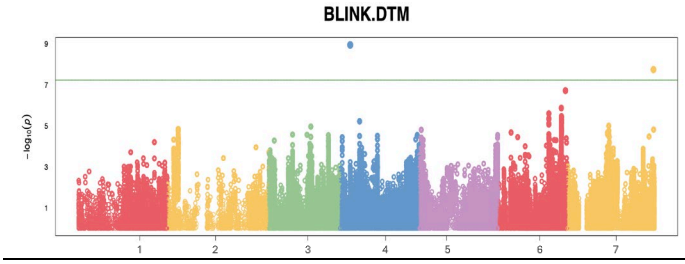


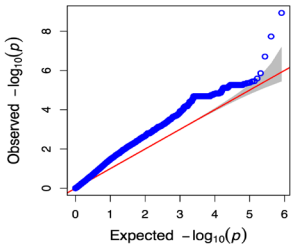


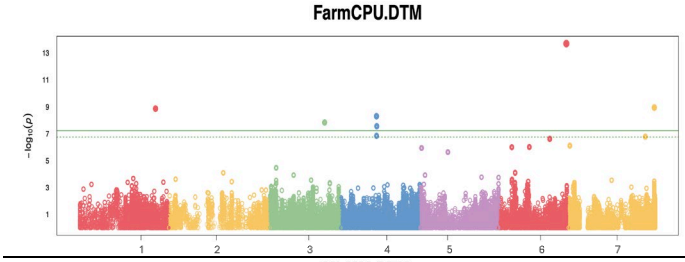
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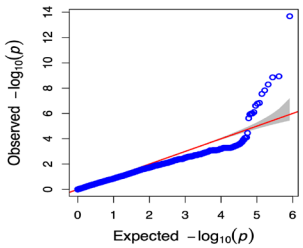


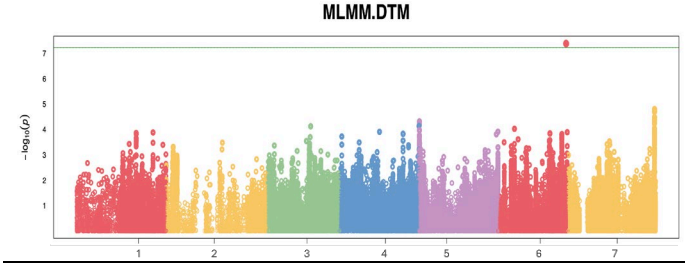
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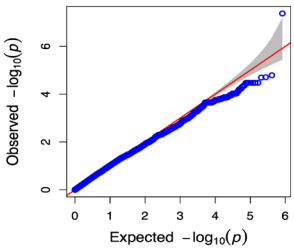






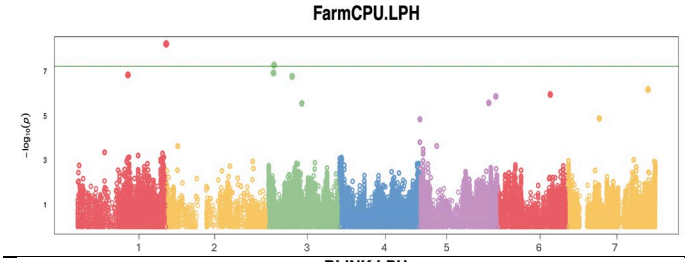


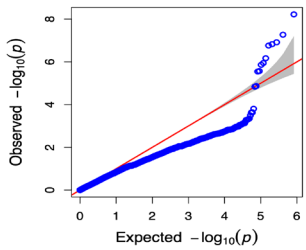


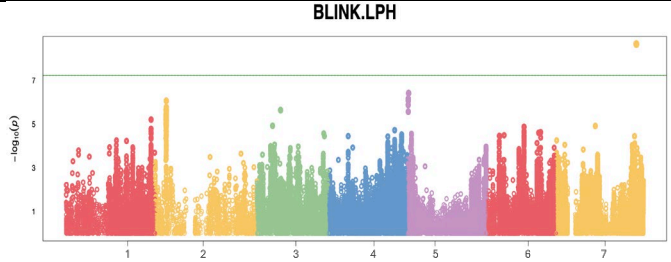


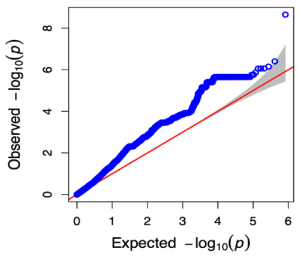
LPH

WPR2022

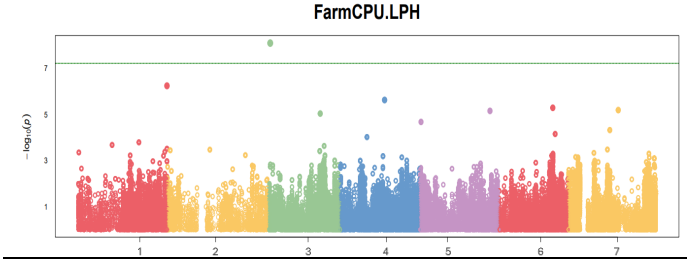


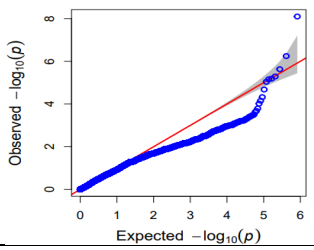




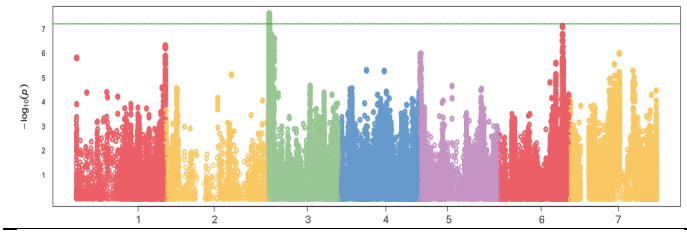


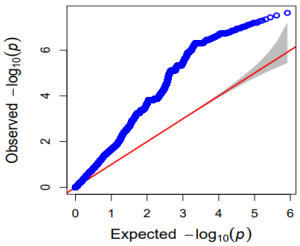
WPR2023



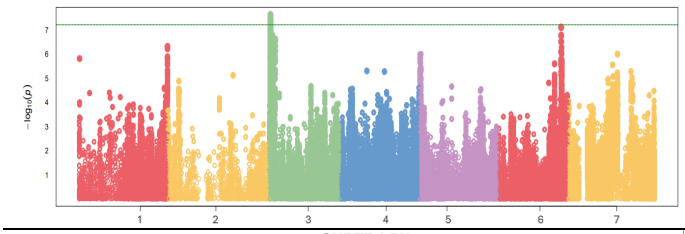


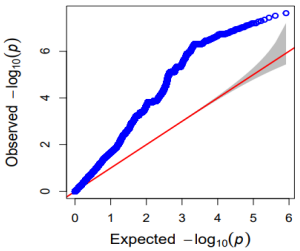
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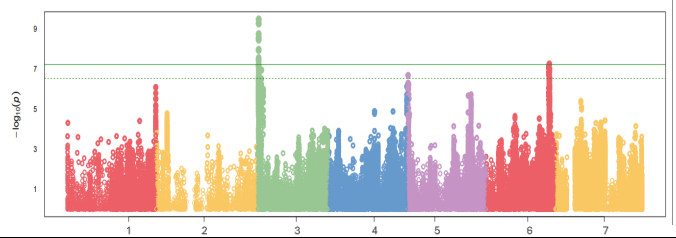


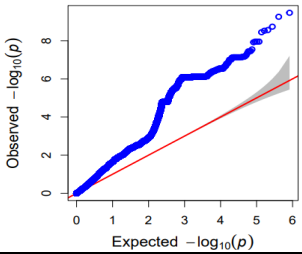
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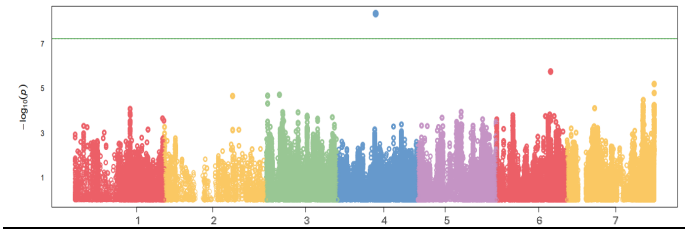
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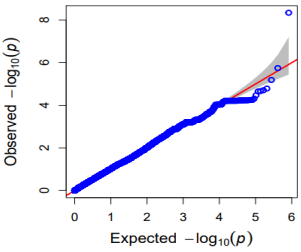




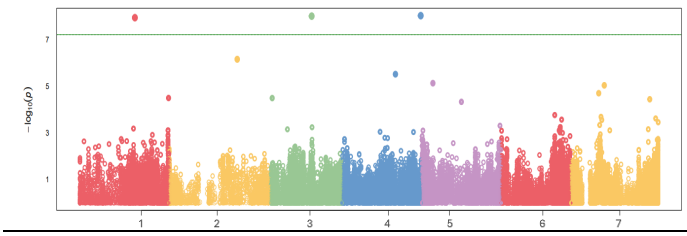
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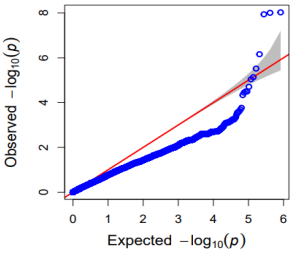
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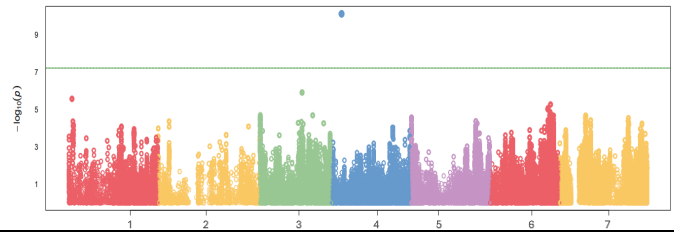
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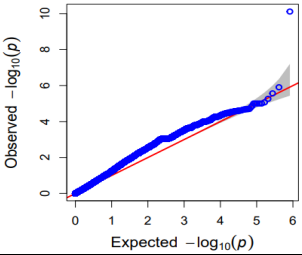




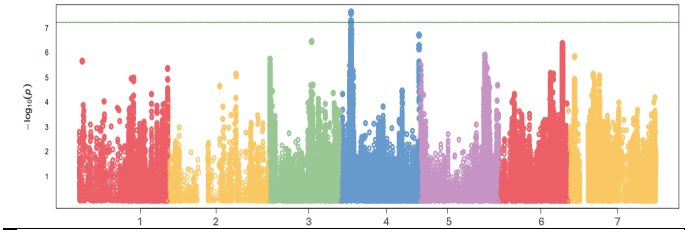
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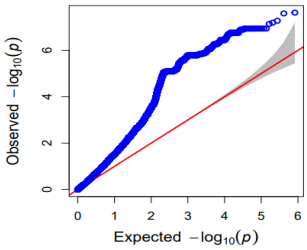
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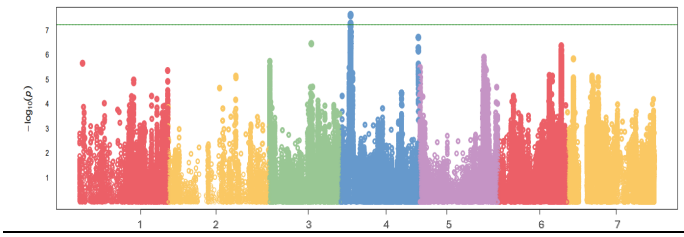


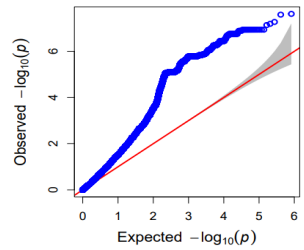
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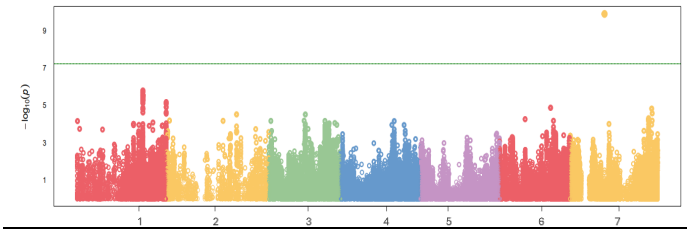


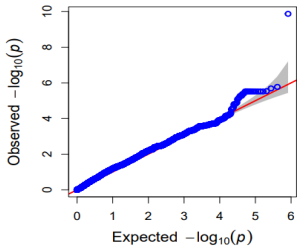


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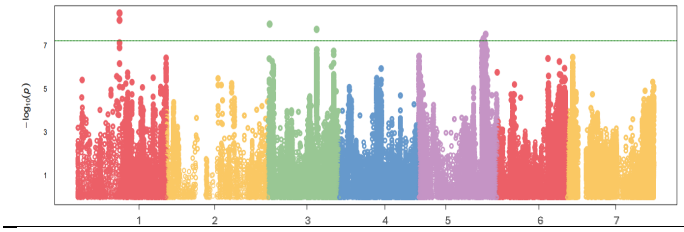
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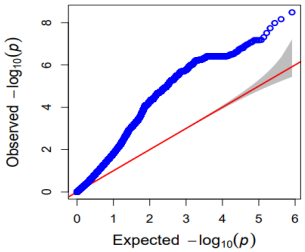




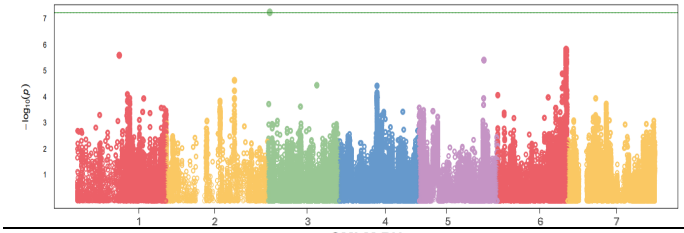
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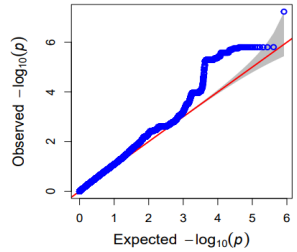
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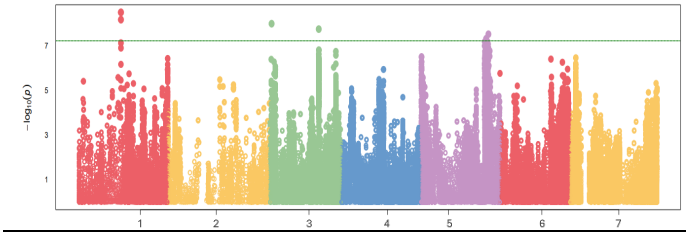


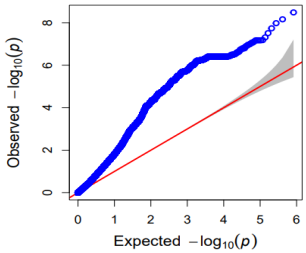
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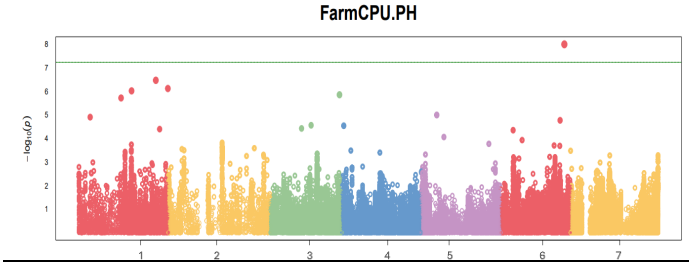


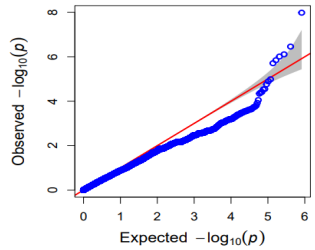
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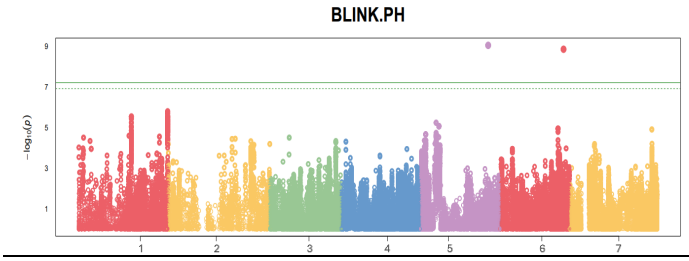


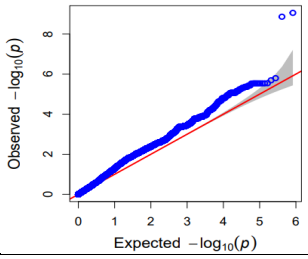


SIM2024



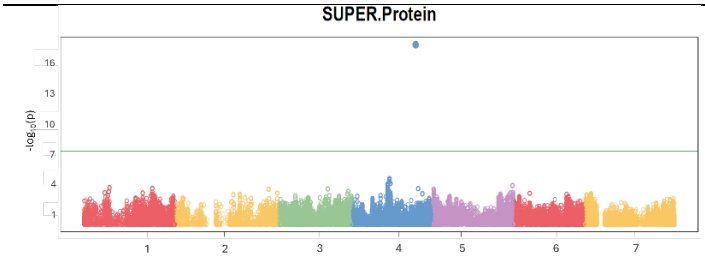


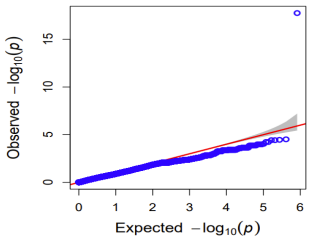




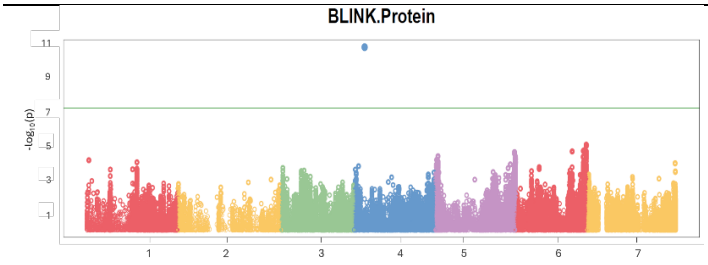
Protein concentration

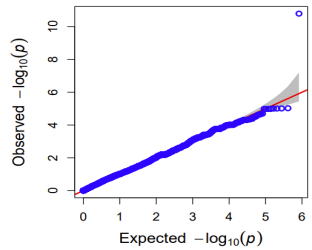
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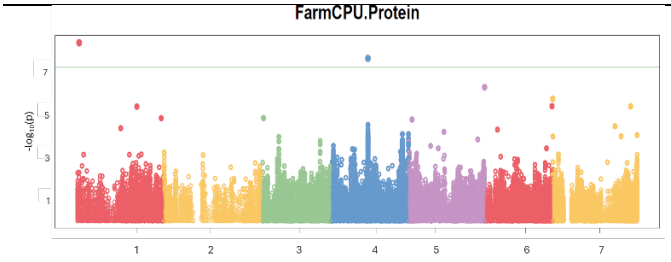


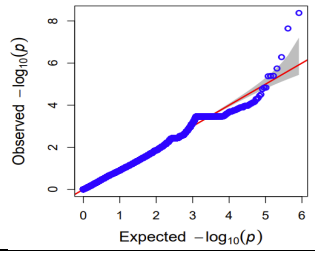


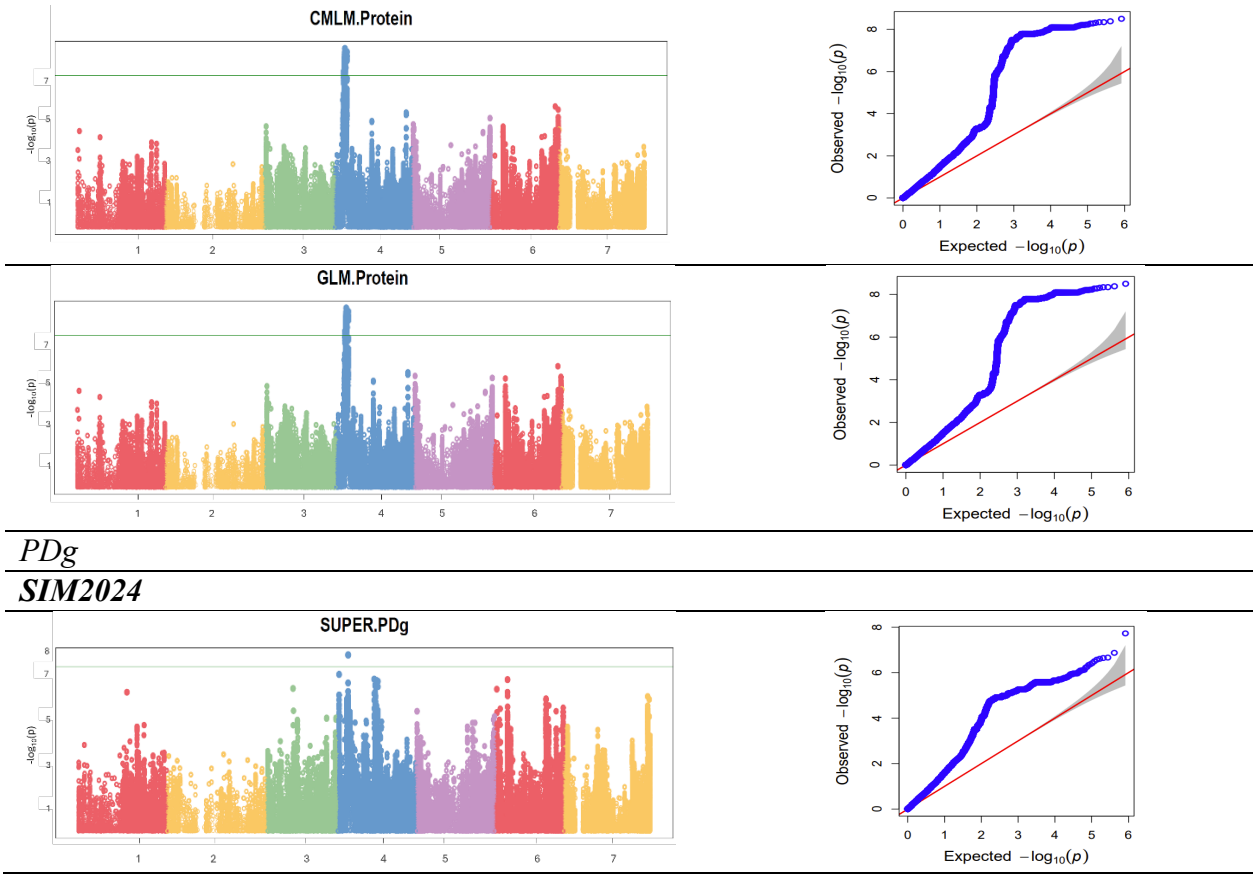
SIM2024











PDg

SIM2024

Figure S3: Manhattan plots and quantile-quantile (QQ) plots with significant SNPs for various traits in the greenhouse and environment-wise multi-environment experiment using LAMP. Bonferroni threshold (p-value < 0.05/ 825,218), indicated by green significant threshold lines, was used to identify significant SNPs.